26

Lecture 26

Future 6G Technologies

فناوری‌های آیندهٔ نسل ششم

2G — 3G — 4G — 5G — **6G**

COURSE PART
Part V – Planes and outlook (Lectures 23–26)

ESTIMATED STUDY TIME
6–8 hours (3 h reading, 3–5 h exercises)

DIFFICULTY
Advanced

PREREQUISITES
Lectures 14, 23 and 25; Shannon capacity and free-space propagation

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Original lecture notes for the course *Advanced Communication Networks*. The lecture sequence follows the course syllabus (26 lectures, 2G to 6G). Equations use identical notation in the English and Persian editions. Companion edition: Persian (`lecture-26-fa.pdf`).

1 Learning objectives

After studying this lecture you should be able to:

- **LO1** **Describe** the IMT-2030 framework: its usage scenarios and overarching aspects.
- **LO2** **Explain** the main candidate technologies: sub-THz spectrum, cell-free massive MIMO, reconfigurable intelligent surfaces, integrated sensing and communication, non-terrestrial networks and AI-native design.
- **LO3** **Compute** free-space path loss at high frequencies and the capacity gain of wider bandwidth under a power limit.
- **LO4** **Estimate** RIS gain, sensing resolution and satellite propagation delay.
- **LO5** **Assess** which claims about 6G are established and which are still research.

2 Prerequisite knowledge

You need the 5G requirements and NR design (Lecture 14), the user-plane budgets of Lecture 23 and the evolution trends of Lecture 25, as well as the Shannon capacity formula and the free-space propagation (Friis) equation.

Key terms. The vocabulary of this lecture with its Persian equivalents:

Term	Persian	Meaning in this lecture
IMT-2030	IMT-2030	ITU framework for the sixth generation (Recommendation M.2160).
Sub-THz	زیرتراهرتز	Carrier frequencies of roughly 100–300 GHz.
Cell-free massive MIMO	انبوه بدون سلول MIMO	Many distributed access points jointly serving all users.
RIS	سطح هوشمند بازپیکربندی‌پذیر	Surface of passive elements with controllable phase that shapes reflections.
ISAC	حسگری و ارتباط یکپارچه	Using the same signals and hardware for radar sensing and communication.
NTN	شبکهٔ غیرزمینی	Satellites and high-altitude platforms as part of the mobile network.
AI-native	هوش‌مصنوعی‌بنیاد	Designed with learned components and data pipelines from the start.
Range resolution	تفکیک‌پذیری فاصله	Smallest distance difference between two distinguishable targets.
Power-limited regime	رژیم محدود به توان	Where more bandwidth no longer increases capacity much.
Energy per bit	انرژی به ازای بیت	Consumed energy divided by delivered bits (J/bit).

3 Introduction and motivation

Each generation has been defined about ten years after the previous one, and 6G follows the same rhythm: the ITU published the IMT-2030 framework in 2023 [2], and 3GPP began 6G study work in Release 20, with normative specifications expected in later releases. As Lecture 25 showed, the next steps in capacity need much wider bandwidths, the cost per bit must keep falling, and large parts of the world are still unconnected. At the same time, the network is expected to do more than move bits: sense its environment, host AI and be sustainable and resilient [6].

Much of 6G is still research. This lecture therefore separates the agreed framework from candidate technologies, and gives simple physical models that let you judge claims about them yourself.

4 Fundamental concepts

4.1 The IMT-2030 framework

Recommendation ITU-R M.2160 [2] defines six usage scenarios (Figure 1). Three extend the 5G triangle: *immersive communication* (from eMBB), *hyper-reliable and low-latency communication* (HRLLC, from URLLC) and *massive communication* (from mMTC). Three are new: *ubiquitous connectivity*, *AI*

and communication and *integrated sensing and communication*. Four overarching aspects apply to all: sustainability, connecting the unconnected, ubiquitous intelligence, and security and resilience. Quantitative requirements are defined in a later ITU step, so numbers such as "1 Tbit/s" found in the literature are research targets, not requirements.

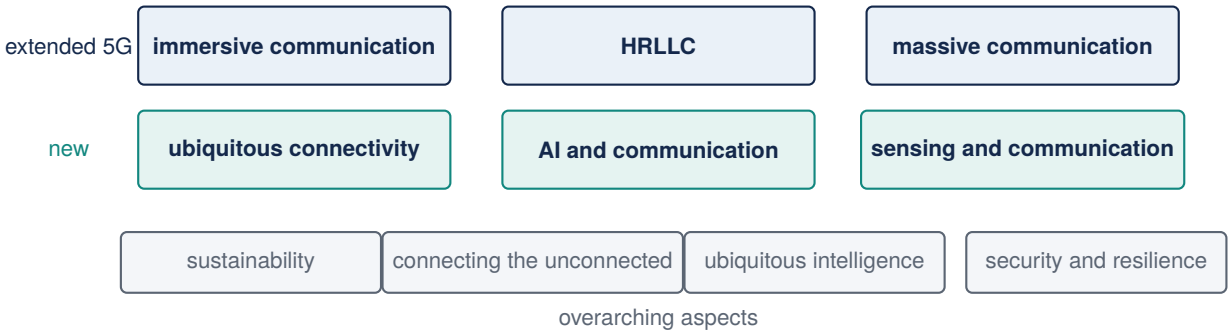


Figure 1. IMT-2030 usage scenarios and overarching aspects (after ITU-R M.2160).

4.2 Candidate technologies

- **New spectrum**: upper mid-band (around 7–24 GHz) for wide-area capacity and sub-THz bands for very high rates at short range [5].
- **Cell-free massive MIMO**: many access points coherently serving users without cell boundaries, giving uniform service [4].
- **Reconfigurable intelligent surfaces**: controllable reflectors that create paths around obstacles at low power [7].
- **Integrated sensing and communication**: the radio as a radar for positioning, mapping and gesture or object detection [3].
- **Non-terrestrial networks**: LEO satellites and high-altitude platforms integrated with the terrestrial system [1].
- **AI-native air interface and network**: learned receivers, beam management and automated operation.

5 Detailed technical explanation

5.1 Sub-THz communication

Above 100 GHz contiguous bandwidths of tens of gigahertz are available, but path loss grows with frequency for antennas of fixed gain, molecular absorption appears at specific frequencies, and blockage by walls, bodies and even foliage is severe. Very large arrays with narrow beams recover the loss (Section 6.1), which makes beam search and tracking the central problem. Hardware is also hard: power amplifiers are inefficient, phase noise grows with frequency, and sampling tens of gigahertz needs low-resolution converters.

5.2 Cell-free networks

In a cellular network users at cell edges suffer from interference and weak signals. In a cell-free network a central processing unit coordinates many distributed access points that all serve all users, with local precoding and fronthaul to the central unit. The result is more uniform rates; the costs are fronthaul and synchronisation – an extension of the CU/DU split and coordination ideas of Lecture 23.

5.3 Reconfigurable intelligent surfaces

An RIS is a panel of many sub-wavelength elements whose reflection phase can be set electronically. Properly configured, it adds a strong reflected path, for example around a blocking building. Its received power grows with the square of the number of elements (Section 6.3), but it suffers the product of two path losses (transmitter to surface, surface to receiver), so it helps most near the transmitter or the receiver.

5.4 Integrated sensing and communication

Wide bandwidths and large arrays are exactly what radars need. ISAC uses communication waveforms or shared hardware to measure range, angle and velocity of objects. Applications include positioning without dedicated infrastructure, traffic monitoring and environment mapping that in turn helps beam management. It also raises privacy questions, since the network could sense people who carry no device.

5.5 Non-terrestrial networks

Release 17 introduced NR over satellite (NTN) with adaptations for large delays and Doppler shifts. LEO constellations offer global coverage with tens of milliseconds of round-trip delay (Section 6.5); integrated with terrestrial networks they address "connecting the unconnected" and resilience after disasters.

5.6 AI-native networks

5G already uses data analytics (NWDAF) and AI in RAN optimisation. 6G aims to make learned components part of the design: AI-based channel state feedback, receivers and beam prediction, and network automation with closed loops. Open questions are standardisation of models, generalisation to unseen conditions, energy cost of the AI itself and explainability.

6 Mathematical models

6.1 Free-space path loss

Model. Between isotropic antennas at distance d and wavelength $\lambda = c/f$:

$$L_{\mathrm{FS}} = 20\log_{10}\left(\frac{4\pi d f}{c}\right)\ \mathrm{dB}. \tag{1}$$

With effective apertures A_t, A_r the Friis equation becomes $P_r = P_t A_t A_r/(\lambda^2 d^2)$, which *grows* with f^2.

Example. At 100 m: 83.3 dB at 3.5 GHz, 101.4 dB at 28 GHz and 115.4 dB at 140 GHz (Figure 2) – 32 dB more at 140 GHz than at 3.5 GHz. Arrays of the same physical size at both ends gain $20\log_{10}(140/3.5) = 32$ dB each, so the link budget can actually improve, at the price of very narrow beams.

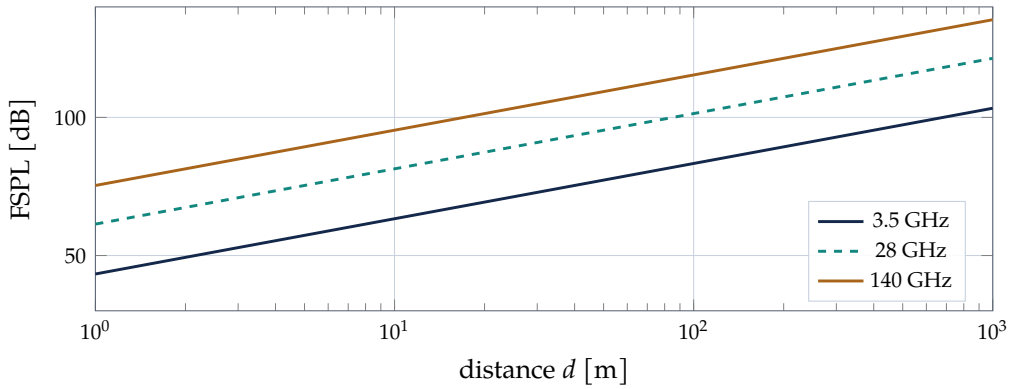


Figure 2. Free-space path loss (Eq. (1)).

6.2 Bandwidth under a power limit

Model. With received power P and noise density N_0:

$$C(B) = B\log_2\left(1+\frac{P}{N_0 B}\right) \xrightarrow{B\to\infty} \frac{P}{N_0 \ln 2}. \tag{2}$$

Example. If the SNR is 20 dB over 1 GHz: $C = 6.66$ Gbit/s; with 10 GHz, 34.6 Gbit/s; with 100 GHz, 100 Gbit/s; the limit is 144 Gbit/s (Figure 3). Wider bandwidth helps greatly only while the SNR stays high – hence the need for array gain.

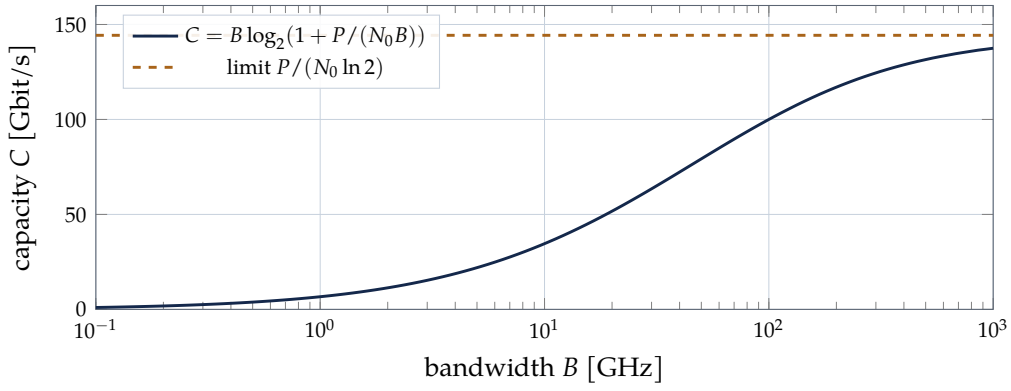


Figure 3. Capacity versus bandwidth at fixed received power (Eq. (2)).

6.3 RIS gain

Model. With N elements whose phases align the reflected signals at the receiver, the amplitudes add coherently:

$$P_r \propto N^2, \qquad G_{\text{RIS}} = 20\log_{10} N \text{ dB} \tag{3}$$

relative to a single element.

Example. $N = 1024$: 60.2 dB over one element (random phases would give only $10\log_{10} N = 30.1$ dB). The gain must overcome the product of the two path losses.

6.4 Sensing resolution

Model. A signal of bandwidth B separates two targets in range if

$$\Delta r = \frac{c}{2B}. \tag{4}$$

Example. $B = 1$ GHz: 15 cm; $B = 10$ GHz: 1.5 cm. Sub-THz bandwidths thus allow centimetre-level sensing.

6.5 Satellite delay

Model. For a satellite at altitude h, Earth radius R and elevation angle ε, the slant range is

$$d(\varepsilon) = \sqrt{(R+h)^2 - R^2\cos^2\varepsilon} - R\sin\varepsilon, \tag{5}$$

and the one-way delay d/c.

Example. $h = 600$ km: 600 km and 2.0 ms at zenith, 1076 km and 3.6 ms at 30°, 1932 km and 6.4 ms at 10° (Figure 4). A geostationary satellite (35 786 km) needs at least 119 ms.

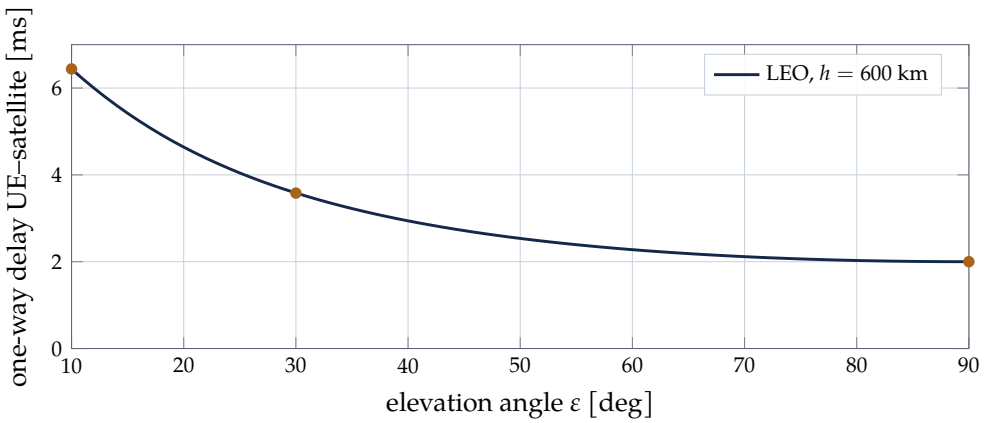


Figure 4. One-way propagation delay to a LEO satellite (Eq. (5)).

7 Network architecture and diagrams

A plausible 6G system keeps the service-based core of 5G (Lecture 18) and extends it with sensing and AI functions, integrates satellite and terrestrial access under one core, and deploys a RAN that mixes macro cells in the upper mid-band, dense sub-THz spots, cell-free clusters and RIS-assisted coverage. Figure 1 gives the service view; Figures 2 to 4 quantify the physical constraints.

8 Step-by-step procedures

Procedure 1 • Checking a sub-THz link budget

1. Compute the free-space loss at the carrier and distance (Eq. (1)); add absorption and blockage margins.
2. Add transmit and receive array gains from the array sizes.
3. Compute the noise power $N_0 B$ including the receiver noise figure, and the SNR.
4. Compute the capacity (Eq. (2)) and compare with the target rate.

Procedure 2 • Judging a 6G claim

1. Is it a requirement in an ITU or 3GPP document or a research target?
2. Which physical quantity limits it (bandwidth, SNR, delay, energy)? Check it with a model of this lecture.
3. What does it cost (spectrum, sites, energy, complexity)?
4. Which usage scenario of Figure 1 does it serve?

9 Worked examples

Worked example 1 • Short-range sub-THz link

Problem. Compute the free-space loss at 300 GHz over 10 m.
Solution. $20\log_{10}(4\pi \times 10 \times 3 \times 10^{11}/3 \times 10^{8}) = 102.0$ dB – about the same as 28 GHz over 100 m, which is why sub-THz suits short ranges.

Worked example 2 • More bandwidth

Problem. With the power of Section 6.2, what capacity does 50 GHz give?
Solution. SNR $= 100/50 = 2$; $C = 50\log_2 3 = 79.2$ Gbit/s: 50 times the bandwidth of the 1 GHz case gives only 12 times the capacity.

Worked example 3 • Satellite round trip

Problem. A UE and a gateway both see a 600 km LEO satellite at 30°. What is the round-trip propagation delay through the satellite (bent pipe)?
Solution. One way UE–satellite–gateway: $2 \times 3.58 = 7.2$ ms; round trip 14.3 ms – compatible with interactive services, unlike GEO (over 477 ms).

10 Technology comparison

Table 1. 5G and candidate 6G features.

	5G (IMT-2020)	6G (IMT-2030, candidates)
Usage scenarios	eMBB, URLLC, mMTC	Immersive, HRLLC, massive, ubiquitous, AI, sensing
Spectrum	Below 7 GHz and 24–71 GHz	+ upper mid-band and sub-THz
Radio architecture	Cells with massive MIMO	Cells + cell-free, RIS
Sensing	Positioning	Integrated sensing and communication
Non-terrestrial	NTN from Release 17	Integrated by design
AI	Analytics (NWDAF), RAN optimisation	AI-native air interface and operation
Status	Deployed	Research and early standardisation

Evolution, not revolution. As every generation before it, 6G will reuse the persistent functions of Lecture 25 and add new capabilities where physics and economics allow.

11 Key takeaways

Key takeaways

- IMT-2030 defines six usage scenarios and four overarching aspects; numerical requirements come later.
- Free-space loss grows as $20\log_{10} f$, but fixed-aperture arrays recover it: beams, not power, are the problem.
- Bandwidth helps only while SNR is high: $C \to P/(N_0 \ln 2)$.
- RIS gain scales as N^2; sensing resolution as $c/(2B)$; LEO delay is a few milliseconds.
- Separate requirements from research targets when reading about 6G.

12 Exercises

Exercise 1 • Level 1 – conceptual • Scenarios

Name the three IMT-2030 usage scenarios that extend 5G and the three new ones.

Exercise 2 • Level 1 – conceptual • Blockage

Give two reasons why sub-THz links are more sensitive to blockage than links at 3.5 GHz, and one mitigation.

Exercise 3 • Level 2 – numerical • Radar resolution

Compute the range resolution of a 400 MHz NR carrier used for sensing.

Exercise 4 • Level 2 – numerical • RIS size

A 16-element RIS gives too weak a reflection. How many elements are needed for 20 dB more received power?

Exercise 5 • Level 3 – analytical • GEO versus LEO

Compute the minimum round-trip delay UE–satellite–gateway for a GEO satellite at 35 786 km and compare with Example 3. Which services are feasible?

Exercise 6 • Level 3 – analytical • Energy

Traffic grows 30 % per year while the network energy budget stays constant. By how much must energy per bit improve per year and over ten years?

Solutions

Solution to Exercise 1

Extended: immersive communication, HRLLC, massive communication. New: ubiquitous connectivity, AI and communication, integrated sensing and communication.

Solution to Exercise 2

Shorter wavelength means less diffraction around obstacles and higher penetration loss through materials; narrow beams are easily blocked. Mitigations: multi-connectivity to several access points, RIS-assisted paths, fast beam switching.

Solution to Exercise 3

$\Delta r = 3 \times 10^8 / (2 \times 4 \times 10^8) = 0.375\,\text{m}$.

Solution to Exercise 4

$(N/16)^2 = 100 \Rightarrow N = 160$ elements.

Solution to Exercise 5

Each hop at least 35 786 km (119.3 ms): one way 238.6 ms, round trip 477 ms, about 33 times the LEO value. Voice is degraded and interactive or real-time control services are not feasible; broadcasting and bulk data work well.

Solution to Exercise 6

Energy per bit must fall by a factor 1.3 each year (23 %), i.e. $1.3^{10} = 13.8$ times in ten years – hence the emphasis on sleep modes, efficient hardware and AI-driven energy saving.

13 Further reading

The IMT-2030 framework is Recommendation ITU-R M.2160 [2]. Saad et al. [6] present a 6G vision; Rappaport et al. [5] discuss frequencies above 100 GHz; Ngo et al. [4] introduce cell-free massive MIMO; Wu and Zhang [7] analyse intelligent reflecting surfaces; Liu et al. [3] survey ISAC; TR 38.811 [1] studies NR for non-terrestrial networks.

References

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